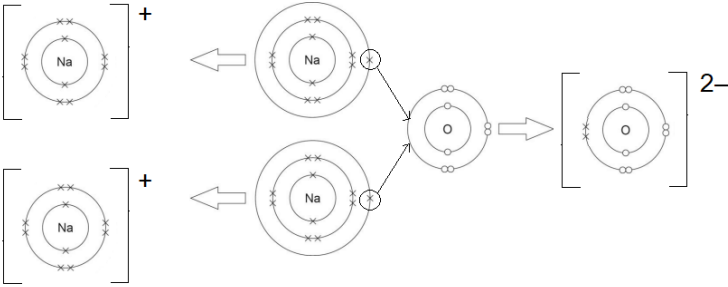
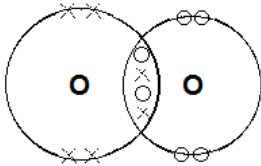


HIGHER TIER ONLY QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		 outer shell electron of both sodium atoms shown going to the outer shell of the oxygen atom (1) correct charges on both sodium ions <u>and</u> the oxide ion (1)		2		2		
		(ii)		 award (2) for correct diagram with two shared pairs of electrons in the overlap <u>and</u> correct octet for both atoms if incorrect award (1) for overlapping oxygen atoms with two shared pairs in the overlap		2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)	I	graphite – (because it has) free/mobile/delocalised electrons (between the layers)	1			1		
			II	graphite – (because) the layers can slide over each other	1			1		
		(ii)		nanotubes / graphene / fullerene / buckminsterfullerene accept 'bucky ball'	1			1		
				Question 2 total	3	4	0	7	0	0